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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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Office Action Summary	Application No. 18/276,688	Applicant(s) VALKONEN, Mika	
	Examiner Jin Ge	Art Unit 2612	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 08/10/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-13 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-13 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 08/10/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Claims 1-13 are pending in the present application.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

Acknowledgment is made of applicant's claim for foreign priority under 35 U.S.C. 119(a)-(d). The certified copy of Finland patent application number FI20215159 filed on 02/17/2021 has been received and made of record.

Information Disclosure Statement

The information disclosure statements (IDS) submitted on 01/13/2025 and 04/25/2024 in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Objections

Claims 1 and 5 objected to because of the following informalities: “analysing” should be “analyzing” (American English).

Claims 9-10 objected to because of the following informalities: “analyse” should be “analyze” (American English). Appropriate correction is required.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP §§ 706.02(l)(1) - 706.02(l)(3) for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims 1-13 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-13 of Application 18276679 in view of Mattioliet et al. (MATTIOLI ET AL; 2D-3D Spatial Registration for Remote Inspection of Power Substations, Faculty of Electrical Engineering, Federal University of Uberlândia, Published 25 November 2020, 24 pages), further in view of WO 2017/191362 to Procemax Oy. This is a provisional nonstatutory double patenting rejection.

Regarding claim 1:

18276688	**18276679**
1. A computer implemented method, comprising: obtaining a previously captured image data of a virtual tour of a location, receiving real-time image data of a target object located within a field of view area of a camera, which target object is	1. A computer implemented method, comprising: obtaining a previously captured image data of a virtual tour of a location, receiving real-time image data of a target object located within a field of view area of a camera, wherein the target

located in the same location wherein the previously captured image data was captured, analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown, receiving real-time acoustic data captured by an audio sensor in the location, analysing the received acoustic data for detecting a malfunction, and wherein the method further comprises: playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for	object -is located in a same location wherein the previously captured image data was captured, overlaying the real-time image data of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour, receiving real-time acoustic data captured by an audio sensor in the location, playing the real-time acoustic data when a user moving along the mixed image virtual tour exceeds a predetermined distance from a position of the target object, wherein the acoustic data is captured by the audio sensor in a real world, and all overlayed image data is captured at a same time point.

further augmenting the mixed image virtual tour by real-time acoustic data.	

Although the conflicting claims are not identical, they are not patentably distinct from each other (see the comparison between claim 1 of the instant invention and claim 1 of the application 18276679) except 1. analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown and 2. analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data.

In related endeavor, Mattioliet et al. teach analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown (abstract, “This work discusses a method to augment virtual environments of power substations with field images, enabling operators to promptly see a virtual representation of the inspected area’s surroundings. In addition, the system interacts with a SCADA database, continuously comparing the equipment states against the ones inferred by processing the field images. Whenever a discrepancy is

found, a virtual camera can be teleported to the affected region, speeding up system reestablishment", Fig 8, section 5.3, "The client evaluates the registration quality in real-time (as explained in Section 6.1), disabling the overlay, whenever the error exceeds a threshold. In this scenario, the overlay is replaced by an icon indicating that condition. With this feature, navigation is not blocked in the augmented virtuality environment. Once the registration becomes poor, due to the high discrepancy between the virtual camera and the photo capture conditions, the overlay is simply disabled, avoiding misinterpretations from the operator. Finally, if the system detects a discrepancy between the state reported by the SCADA database and the state inferred from the image, an alarm dialog is presented to the user (Figure 9). The alarms can be either ignored (Dismiss button) or iconified (OK button). Alternatively, the dialog gives an option to teleport to the affected asset, for further inspection and contextualization. In this case, the environment's viewport is changed so that the photo is overlaid in the 3D model, leaving some space on the screen edges to see the surroundings, using the camera parameters stored in the database.").

In related endeavor, Procemax Oy teaches analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data (page 2, lines 13-37, "transmitting recorded acoustic data to the data processing device, forming a second audio profile for acoustic data recorded by the second sound acquisition unit, comparing the second audio profile to pre-stored operational state audio profiles, and if at least the first and the

second audio profiles correspond to an audio profile of a malfunction operational state and comprise an abnormal noise: locating a malfunction of the process machinery on the basis of locations of the first and the second sound acquisition units, and intensity of abnormal noise in the first and the second audio profiles ", page 8, lines 22-36, "If at least two sound acquisition units of a plurality of sound acquisition units arranged in the process machinery environment transmits data where the acoustic spectrum has the same or at least partly the same kind of deviation from a normal state acoustic spectrum, it is possible to locate the source that causes the deviation on the basis of intensity of the deviation of each acoustic spectrum").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified claim 1 of the 18276679 to include analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown as taught by Mattioliet et al. to augment virtual environments of power substations with field images, enabling operators to promptly see a virtual representation of the inspected area's surroundings to provide the registration accuracy and performance impact for a simple scenario.

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified claim 1 of the 18276679 with Mattioliet et al. to include analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed

image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data as taught by Procemax Oy to combine the malfunction is detected by real-time acoustic data, as taught by Procemax Oy, with playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location, as taught by claim 1 of Application 18277679 to playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data to acoustically monitoring the operational state of process machinery through an acoustic sensor to determine malfunction status of the process machinery.

Likewise instant dependent claims 2-8 and 13 are anticipated by dependent claims 2-8 and 13 of the application 18276679, and is not patentably distinct from claims 2-8 and 13 of the application 18276679.

Regarding claim 9:

18276688	**18276679**
9. A camera system comprising at least one camera and a data processing unit, wherein the data processing unit is configured to obtain a previously captured image data of a virtual tour of a location, and which the at least one camera is	9. A system comprising at least one camera and a data processing unit, wherein the data processing unit is configured to obtain a previously captured image data of a virtual tour of a location and the at least one camera is configured

configured to capture real-time image data of a target object located within a field of view area of the at least one camera, and which target object is located in the same location wherein the previously captured image data was captured; and wherein the data processing unit is further configured to analyse the real-time image data for detecting a deviation and overlay the real-time image data of the target object on a corresponding area of a virtual tour view for forming a mixed image virtual tour, only if the deviation is detected, otherwise the virtual tour view of the target object is shown, and which the system further comprises an audio sensor configured to capture acoustic data in the location and to transmit the captured acoustic data to the data processing unit for analysis, wherein if a malfunction is detected, the real-time acoustic data is configured to be played in an area of the mixed image	to capture real-time image data of a target object located within a field of view area of the at least one camera, and wherein the target object is located in a same location where the captured image data was captured, and to transmit the captured image data to the data processing unit, and wherein the data processing unit is further configured to overlay the real-time image data of the target object on a corresponding area of the virtual tour view for forming a mixed image virtual tour, to receive real-time acoustic data captured by an audio sensor in the location, and to play the real-time acoustic data when the user moving through the mixed image virtual tour exceeds a predetermined distance from a position of the target object, the acoustic data being captured the audio sensor, and all overlayed image data is captured at a same time point.

virtual tour view corresponding to the location of the audio sensor or its surroundings in a real environment.	

Although the conflicting claims are not identical, they are not patentably distinct from each other (see the comparison between claim 9 of the instant invention and claim 9 of the application 18276679) except 1. analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown and 2. analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data.

In related endeavor, Mattioliet et al. teach analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown (abstract, “This work discusses a method to augment virtual environments of power substations with field images, enabling operators to promptly see a virtual representation of the inspected area’s surroundings. In addition, the system interacts with a SCADA database, continuously comparing the equipment states

against the ones inferred by processing the field images. Whenever a discrepancy is found, a virtual camera can be teleported to the affected region, speeding up system reestablishment", Fig 8, section 5.3, "The client evaluates the registration quality in real-time (as explained in Section 6.1), disabling the overlay, whenever the error exceeds a threshold. In this scenario, the overlay is replaced by an icon indicating that condition. With this feature, navigation is not blocked in the augmented virtuality environment. Once the registration becomes poor, due to the high discrepancy between the virtual camera and the photo capture conditions, the overlay is simply disabled, avoiding misinterpretations from the operator. Finally, if the system detects a discrepancy between the state reported by the SCADA database and the state inferred from the image, an alarm dialog is presented to the user (Figure 9). The alarms can be either ignored (Dismiss button) or iconified (OK button). Alternatively, the dialog gives an option to teleport to the affected asset, for further inspection and contextualization. In this case, the environment's viewport is changed so that the photo is overlaid in the 3D model, leaving some space on the screen edges to see the surroundings, using the camera parameters stored in the database.").

In related endeavor, Procemax Oy teaches analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data (page 2, lines 13-37, "transmitting recorded acoustic data to the data processing device, forming a second audio profile for acoustic data recorded by the second sound acquisition unit, comparing the second

audio profile to pre-stored operational state audio profiles, and if at least the first and the second audio profiles correspond to an audio profile of a malfunction operational state and comprise an abnormal noise: locating a malfunction of the process machinery on the basis of locations of the first and the second sound acquisition units, and intensity of abnormal noise in the first and the second audio profiles ", page 8, lines 22-36, "If at least two sound acquisition units of a plurality of sound acquisition units arranged in the process machinery environment transmits data where the acoustic spectrum has the same or at least partly the same kind of deviation from a normal state acoustic spectrum, it is possible to locate the source that causes the deviation on the basis of intensity of the deviation of each acoustic spectrum").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified claim 9 of the 18276679 to include analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown as taught by Mattioliet et al. to augment virtual environments of power substations with field images, enabling operators to promptly see a virtual representation of the inspected area's surroundings to provide the registration accuracy and performance impact for a simple scenario.

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified claim 1 of the 18276679 with Mattioliet et al. to include analysing the received acoustic data for

detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data as taught by Procemax Oy to combine the malfunction is detected by real-time acoustic data, as taught by Procemax Oy, with playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location, as taught by claim 1 of Application 18277679 to playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data to acoustically monitoring the operational state of process machinery through an acoustic sensor to determine malfunction status of the process machinery.

Likewise instant dependent claims 11-12 are anticipated by dependent claims 11-12 of the application 18276679, and is not patentably distinct from claims 11-12 of the application 18276679.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claim(s) 1-2, 4-7, 9-11, and 13 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. PGPubs 2016/0127690 to Kaehler et al. in view of Mattioliet et al. (MATTIOLI ET AL; 2D-3D Spatial Registration for Remote Inspection of Power Substations, Faculty of Electrical Engineering, Federal University of Uberlândia, Published 25 November 2020, 24 pages), further in view of U.S. PGPubs 2018/0115759 to Sawa et al., further in view of WO 2017/191362 to Procemax Oy.

Regarding claim 1, Kaehler et al. teach **a computer implemented method** (par 0004), **comprising:**

obtaining a previously captured image data of a virtual tour of a location (par 0003, "A memory stores data associated with a rendered three-dimensional virtual environment associated with a geographic region that includes the region of interest", par 0019, " In the example of FIG. 1, the area monitoring system **10** also includes a virtual environment **14**, such as stored in memory. The virtual environment **14** can correspond to a rendered three-dimensional virtual environment that is associated with substantially the entire geographic region that includes at least the respective region(s) of interest associated with the camera(s) **12"**),

receiving real-time image data of a target object located within a field of view area of a camera, which target object is located in the same location wherein the previously captured image data was captured (par 0016, "The user interface can include a display that is configured to display the image data at a location perspective corresponding to a viewing perspective of the user at a given virtual location

within the rendered three-dimensional virtual environment …the user can view the real-time video images provided via the camera(s) based on the location perspective of the user in the virtual environment relative to the viewing perspective of the camera", par 0018, " The area monitoring system **10** includes at least one camera **12** that is configured to monitor a region of interest of a geographic region and to provide real-time video data corresponding to real-time video images RTVI of the region of interest"), **and**

overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data (par 0003, "An image processor combines the real-time video data and the data associated with the three-dimensional virtual environment to generate image data. A user interface enables a user to navigate the three-dimensional virtual environment via inputs and displays the image data based on a location perspective corresponding to a viewing perspective of the user at a given virtual location within the three-dimensional virtual environment. The image processor superimposes the video images of the region of interest onto the three-dimensional virtual environment in the image data at a location and an orientation based on the location perspective of the user within the three-dimensional virtual environment", par 0016, "The user interface can include a display that is configured to display the image data at a location perspective corresponding to a viewing perspective of the user at a given virtual location within the rendered three-dimensional virtual environment. Thus, the displayed image data presents the location perspective as a virtual location of the user in the virtual environment at a viewing perspective corresponding to the virtual location and viewing orientation of the user in the virtual environment. Additionally, the image processor can

be further configured to superimpose the real-time video images of the region of interest onto the virtual environment in the image data at an orientation associated with the location perspective of the user within the rendered three-dimensional virtual environment. As a result, the user can view the real-time video images provided via the camera(s) based on the location perspective of the user in the virtual environment relative to the viewing perspective of the camera", par 0021-0023, "Based on the combination of the real-time video data that is provided via the camera(s) **12** with the virtual environment **14**, the image processor **16** can be further configured to superimpose the real-time video images RTVI of the region of interest from the camera(s) **12** onto the virtual environment **14** in the image data at an orientation associated with the location perspective of the user within the virtual environment **14**.").

But Kaehler et al. keep silent for teaching **analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown**.

In related endeavor, Mattioliet et al. teach **analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown** (abstract, "This work discusses a method to augment virtual environments of power substations with field images, enabling operators to promptly see a virtual representation of the inspected

area's surroundings. In addition, the system interacts with a SCADA database, continuously comparing the equipment states against the ones inferred by processing the field images. Whenever a discrepancy is found, a virtual camera can be teleported to the affected region, speeding up system reestablishment", Fig 8, section 5.3, "The client evaluates the registration quality in real-time (as explained in Section 6.1), disabling the overlay, whenever the error exceeds a threshold. In this scenario, the overlay is replaced by an icon indicating that condition. With this feature, navigation is not blocked in the augmented virtuality environment. Once the registration becomes poor, due to the high discrepancy between the virtual camera and the photo capture conditions, the overlay is simply disabled, avoiding misinterpretations from the operator. Finally, if the system detects a discrepancy between the state reported by the SCADA database and the state inferred from the image, an alarm dialog is presented to the user (Figure 9). The alarms can be either ignored (Dismiss button) or iconified (OK button). Alternatively, the dialog gives an option to teleport to the affected asset, for further inspection and contextualization. In this case, the environment's viewport is changed so that the photo is overlaid in the 3D model, leaving some space on the screen edges to see the surroundings, using the camera parameters stored in the database.").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified Kaehler et al. to include analysing the real-time image data for detecting a deviation, and overlaying the real-time image data on of the target object on a corresponding area of the target object in a virtual tour view for forming a mixed image virtual tour data, only if the deviation is detected, otherwise the virtual tour view of the target object is shown as taught by

Mattioliet et al. to augment virtual environments of power substations with field images, enabling operators to promptly see a virtual representation of the inspected area's surroundings to provide the registration accuracy and performance impact for a simple scenario.

But Kaehler et al. as modified by Mattioliet et al. do not explicitly teach **receiving real-time acoustic data captured by an audio sensor in the location, analysing the received acoustic data for detecting a malfunction, and wherein the method further comprises: playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data**.

In related endeavor, Sawa et al. teach **receiving real-time acoustic data captured by an audio sensor in the location** (par 0083, " the microphone array **20** starts to collect sound of the vicinity of the monitoring target point (location) (S**2**)", par 0132, "that is, videos are captured and sound is collected in real time"), **and wherein the method further comprises: playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location** (par 0012, "outputting the recorded audio data from a sound output unit as sound; receiving designation of one or more designated locations in the video data displayed on the display unit; and generating or combining audio data in which sound is emphasized in directivities directed toward positions corresponding to the one or more designated locations in the video data from the sound collecting unit based on the recorded audio data", par 0091, "The reproducing

unit **60** outputs, as sound, the respective audio data items in which the sound (volume level thereof) is emphasized (amplified) in the directions directed toward the respective sound positions corresponding to one or more designated locations or the sound emphasis region by the signal processing unit **50**, from the speaker **65** in synchronization with the video data which is being displayed on the display **63** in response to the designation in step S**11** (S**14**)").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified Kaehler et al. as modified by Mattioliet et al. Sawa et al. to include receiving real-time acoustic data captured by an audio sensor in the location, and wherein the method further comprises: playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location as taught by Sawa et al. to generate or combine audio data in which sound is emphasized in directivities directed toward positions corresponding to the one or more designated locations in the video data from the sound collecting unit based on the audio data recorded in the recording unit.

In related endeavor, Procemax Oy teaches analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data (page 2, lines 13-37, "transmitting recorded acoustic data to the data processing device, forming a second audio profile for acoustic data recorded by the second sound acquisition unit, comparing the second

audio profile to pre-stored operational state audio profiles, and if at least the first and the second audio profiles correspond to an audio profile of a malfunction operational state and comprise an abnormal noise: locating a malfunction of the process machinery on the basis of locations of the first and the second sound acquisition units, and intensity of abnormal noise in the first and the second audio profiles ", page 8, lines 22-36, "If at least two sound acquisition units of a plurality of sound acquisition units arranged in the process machinery environment transmits data where the acoustic spectrum has the same or at least partly the same kind of deviation from a normal state acoustic spectrum, it is possible to locate the source that causes the deviation on the basis of intensity of the deviation of each acoustic spectrum").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified Kaehler et al. as modified by Mattioliet et al. and Sawa et al. to include analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data as taught by Procemax Oy to combine the malfunction is detected by real-time acoustic data, as taught by Procemax Oy, with playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location, as taught by Sawa et al. to playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the

mixed image virtual tour by real-time acoustic data to acoustically monitoring the operational state of process machinery through an acoustic sensor to determine malfunction status of the process machinery.

Regarding claim 2, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, and Kaehler et al. further teach wherein the previously captured image data is static image data of a real environment of the location (par 0035, "FIG. 3 illustrates a first example of image data **100**. As described previously, the image data **100** can correspond to the image data IMG that is provided to the user interface **62** and which is displayed to the user via the display **64** in the example of FIG. 2. The image data **100** is provided in the example of FIG. 3 as an overhead view of a portion of the virtual environment **58**, and thus of a portion of the geographic region (e.g., the airport)").

Regarding claim 4, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, and Kaehler et al. further teach wherein the previously captured image data is a ready-constructed virtual tour (par 0003, "A memory stores data associated with a rendered three-dimensional virtual environment associated with a geographic region that includes the region of interest. An image processor combines the real-time video data and the data associated with the three-dimensional virtual environment to generate image data. A user interface enables a user to navigate the three-dimensional virtual environment via inputs and displays the image data based on a location perspective corresponding to a viewing perspective of

the user at a given virtual location within the three-dimensional virtual environment", par 0035, "The image data **100** includes a compass rose **102** and a set of controls **104** that can assist the user in navigating through the virtual environment **58**. The set of controls **104** includes direction arrows **106** to allow the user to pan over the overhead view of the virtual environment **58**, such as to explore other portions of the virtual environment **58** corresponding to respective other portions of the geographic region").

Regarding claim 5, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, and further teach wherein the method further comprises: obtaining additional information about the location (Kaehler et al.: par 0037, "the image data **100** includes a plurality of dynamic object icons **126** that are demonstrated at virtual locations that can correspond to respective approximate three-dimensional locations in the geographic region. In the example of FIG. 3, the dynamic object icons **126** can correspond to airplanes located in and/or moving along various portions of the geographic region"), analysing the additional information for detecting a deviation (Mattioliet et al.: Fig 8, section 5.3, "The client evaluates the registration quality in real-time (as explained in Section 6.1), disabling the overlay, whenever the error exceeds a threshold. In this scenario, the overlay is replaced by an icon indicating that condition. With this feature, navigation is not blocked in the augmented virtuality environment. Once the registration becomes poor, due to the high discrepancy between the virtual camera and the photo capture conditions, the overlay is simply disabled, avoiding misinterpretations from the operator. Finally, if the system detects a discrepancy between the state reported by the SCADA database and

the state inferred from the image, an alarm dialog is presented to the user (Figure 9). The alarms can be either ignored (Dismiss button) or iconified (OK button). Alternatively, the dialog gives an option to teleport to the affected asset, for further inspection and contextualization. In this case, the environment's viewport is changed so that the photo is overlaid in the 3D model, leaving some space on the screen edges to see the surroundings, using the camera parameters stored in the database."), and overlaying the additional information on the mixed image virtual tour view if a deviation is detected (Kaehler et al.: par 0037, "the location of the dynamic object icons **126** on the virtual environment **58** in the image data **100** can be based on the received dynamic object inputs DO_DT (e.g., based on GPS inputs). Therefore, the image controller **60** can be configured to update the virtual location of the dynamic object icons **126** in substantial real-time in response to movement of the dynamic objects in the geographic region", Mattioliet et al.: Fig 7, section 5.3, "the VR client query the list of field cameras from the server. A red marker is added above the virtual instances of the monitored assets, indicating the initial state (no registration), as shown in Figure 7.", Fig 8, section 5.3, "The client evaluates the registration quality in real-time (as explained in Section 6.1), disabling the overlay, whenever the error exceeds a threshold. In this scenario, the overlay is replaced by an icon indicating that condition. With this feature, navigation is not blocked in the augmented virtuality environment").

Regarding claim 6, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, and Kaehler et al. further teach wherein the real-time image data is real-time video stream or frequently captured real-

time images (par 0003, "The system includes a camera to monitor a region of interest and provide real-time video data corresponding to real-time video images of the region of interest").

Regarding claim 7, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, and Kaehler et al. further teach wherein the method further comprises: capturing real-time image data configured to be overlayed on areas of the virtual tour view by more than one cameras (par 0004, "The method includes receiving real-time video data corresponding to real-time video images of a region of interest within the geographic region via a plurality of cameras .... The method further includes superimposing the real-time video images of the region of interest associated with at least one of the plurality of cameras in the location perspective of the user onto the rendered three-dimensional virtual environment in the image data at a location and an orientation based on the location perspective of the user within the rendered three-dimensional virtual environment", Fig. 2, par 0025, "In the example of FIG. 2, the area monitoring system **50** includes a plurality X of cameras **52**, where X is a positive integer. The cameras **52** are each configured to monitor a region of interest of a geographic region and to provide respective real-time video data streams VID.sub.1 through VID.sub.X corresponding to real-time video images RTVI.sub.1 through RTVI.sub.X of the region of interest").

Regarding claim 9, Kaehler et al. teach a camera system comprising at least one camera (Fig. 2, par 0025, "In the example of FIG. 2, the area monitoring

system **50** includes a plurality X of cameras **52**, where X is a positive integer. The cameras **52** are each configured to monitor a region of interest of a geographic region and to provide respective real-time video data streams VID.sub.1 through VID.sub.X corresponding to real-time video images RTVI.sub.1 through RTVI.sub.X of the region of interest") and a data processing unit (par 0027, "The area monitoring system **50** can also include an image processor **60** that is configured to combine the processed real-time video data, demonstrated in the example of FIG. 2 as a signal VID_DT, with the virtual environment data **58**, demonstrated in the example of FIG. 2 as a signal VE, to generate image data IMG that is provided to a user interface **62** and with which a user can interact"). The remaining limitations of the claim are similar in scope to claim 1 and rejected under the same rationale.

Regarding claim 10, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 9, the claim 10 is similar in scope to claim 5 and is rejected under the same rational.

Regarding claim 11, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 9, the claim 11 is similar in scope to claim 7 and is rejected under the same rational.

Regarding claim 13, Kaehler et al. teach a computer program product, stored on a non-transitory computer readable medium and executable in a computing device, wherein the computer program product comprises instructions to perform the method

according to claim 1 (par 0004). The remaining limitations of the claim are similar in scope to claim 1 and rejected under the same rationale.

Claim(s) 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. PGPubs 2016/0127690 to Kaehler et al. in view of Mattioliet et al. (MATTIOLI ET AL; 2D-3D Spatial Registration for Remote Inspection of Power Substations, Faculty of Electrical Engineering, Federal University of Uberlândia, Published 25 November 2020, 24 pages), further in view of U.S. PGPubs 2018/0115759 to Sawa et al., further in view of WO 2017/191362 to Procemax Oy, further in view of WO2020/069049 to Gausebeck et al..

Regarding claim 3, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, and Kaehler et al. further teach which previously captured image data is obtained from a camera, a memory or a cloud storage, and wherein the method further comprises: constructing the previously captured image data as a virtual tour by a data processing unit (par 0003, " The system includes a camera to monitor a region of interest and provide real-time video data corresponding to real-time video images of the region of interest. A memory stores data associated with a rendered three-dimensional virtual environment associated with a geographic region that includes the region of interest. An image processor combines the real-time video data and the data associated with the three-dimensional virtual environment to generate image data. A user interface enables a user to navigate the three-dimensional virtual environment via inputs and displays the image data based on

a location perspective corresponding to a viewing perspective of the user at a given virtual location within the three-dimensional virtual environment"), but keep silent for teaching wherein the previously captured image data comprises images captured by a person walking through the location.

In related endeavor, Gausebeck et al. teaches wherein the previously captured image data comprises images captured by a person walking through the location (par 0084-0085, "2D image data of the portion of the house depicted in the 3D floorplan model was captured by a camera held and operated by a user as the user walked from room to room and took pictures of the house from different perspectives within the rooms (e.g., while standing on the floor).").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy to include wherein the previously captured image data comprises images captured by a person walking through the location as taught by Gausebeck et al. to generating 3D data for 2D images using affordable, user friendly devices and techniques for accurately and efficiently aligning the 2D images using the 3D data to generate immersive 3D environments are in high demand to allow users to view multiple 360-degree (360°) panoramas through rendering multiple captured image around environment.

Claim(s) 8 and 12 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. PGPubs 2016/0127690 to Kaehler et al. in view of Mattioliet et al. (MATTIOLI ET AL; 2D-3D Spatial Registration for Remote Inspection of Power

Substations, Faculty of Electrical Engineering, Federal University of Uberlândia, Published 25 November 2020, 24 pages), further in view of U.S. PGPubs 2018/0115759 to Sawa et al., further in view of WO 2017/191362 to Procemax Oy, further in view of U.S. PGPubs 2020/0322917 to Rosenberg et al..

Regarding claim 8, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 1, but keep silent for teaching wherein a time point of a view of the mixed image virtual tour is changeable to a previous time point.

In related endeavor, Rosenberg et al. teaches wherein a time point of a view of the mixed image virtual tour is changeable to a previous time point (par 0122, "Such timestamped data can include video, audio, and/or location data captured by IP device **22** during the walkthrough. Further, timestamped data may be reported to ESW server end **26** on a repeated basis and then stored by the ESW host service in database **68** as part of a record or log of the walkthrough. With respect to location data, such data can be utilized to construct a timeline of the location of IP device **22** during the enhanced structure walkthrough. Thus, such a walkthrough timeline may indicate the duration of the walkthrough, the amount of time the interested party spent in different regions of the structure (e.g., rooms of the single family home) or the structure's exterior area, and the like").

It would have been obvious to a person of ordinary skill in the art at the time before the effective filing data of the claimed invention to modified Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy to include wherein a time

point of a view of the mixed image virtual tour is changeable to a previous time point as taught by Rosenberg et al. to allow user to review back the structure associated with various drawbacks based on the timeline improve to the overall experience of an interested party.

Regarding claim 12, Kaehler et al. as modified by Mattioliet et al., Sawa et al., and Procemax Oy teach all the limitation of claim 9, the claim 12 is similar in scope to claim 8 and is rejected under the same rational.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Jin Ge whose telephone number is (571)272-5556. The examiner can normally be reached 8:00 to 5:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jason Chan can be reached at (571)272-3022. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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